

Problem

A balanced three-phase generator has an *abc* phase sequence with phase voltage

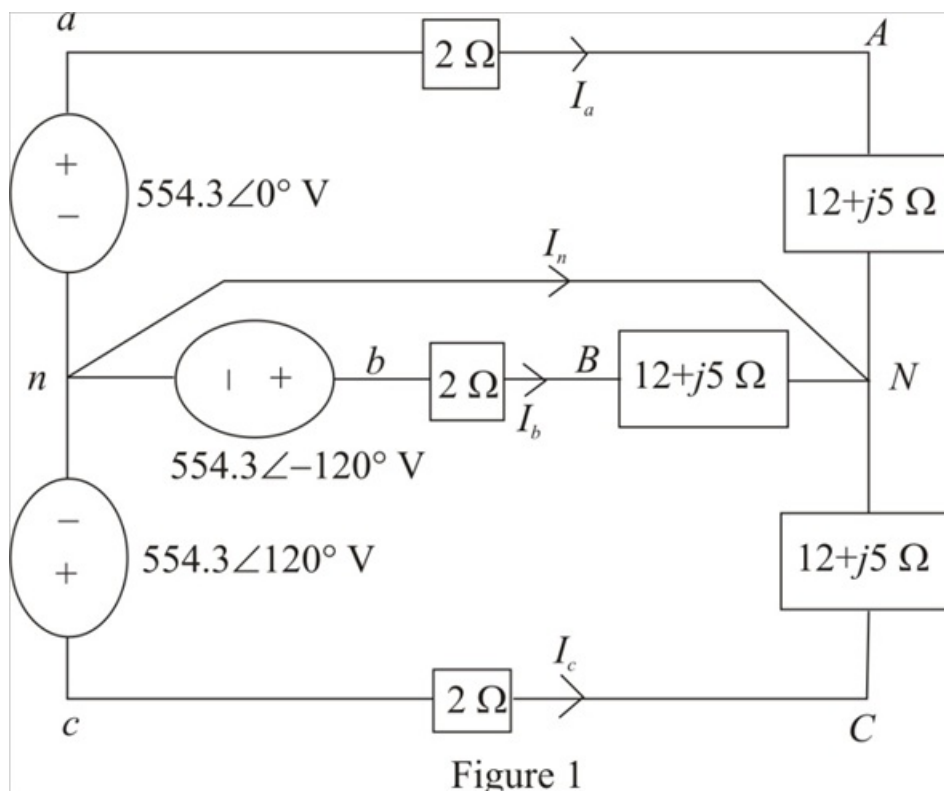
$\underline{V}_{an} = 554.3 \angle 0^\circ \text{ V}$. The generator feeds an induction motor which may be

represented by a balanced Y-connected load with an impedance of $12 + j5 \Omega$ per phase. Find the line currents and the load voltages. Assume a line impedance of 2Ω per phase.

Step-by-step solution

Step 1 of 7

Consider the circuit diagram shown in Figure 1.



Step 2 of 7

Calculate the value of the current I_a .

$$\begin{aligned}
 I_a &= \frac{554.3 \angle 0^\circ}{2 + 12 + j5} \\
 &= \frac{554.3 \angle 0^\circ}{14 + j5} \\
 &= \frac{554.3 \angle 0^\circ}{14.87 \angle 19.65^\circ} \\
 &= 37.28 \angle -19.65^\circ \text{ A}
 \end{aligned}$$

Therefore, the value of the current I_a is, $37.26 \angle -19.65^\circ \text{ A}$.

Step 3 of 7

Calculate the value of the current I_b .

$$\begin{aligned} I_b &= \frac{554.3\angle -120^\circ}{2+12+j5} \\ &= \frac{554.3\angle -120^\circ}{14+j5} \\ &= \frac{554.3\angle -120^\circ}{14.87\angle 19.65^\circ} \\ &= 37.28\angle -139.65^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_b is, $37.28\angle -139.65^\circ \text{ A}$.

Step 4 of 7

Calculate the value of the current I_c .

$$\begin{aligned} I_c &= \frac{554.3\angle 120^\circ}{2+12+j5} \\ &= \frac{554.3\angle 120^\circ}{14+j5} \\ &= \frac{554.3\angle 120^\circ}{14.87\angle 19.65^\circ} \\ &= 37.28\angle 100.35^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_c is, $37.28\angle 100.35^\circ \text{ A}$.

Step 5 of 7

Calculate the value of the voltage V_{AN} .

$$V_{AN} = I_a(12+j5)$$

Substitute $37.28\angle -19.65^\circ \text{ A}$ for I_a .

$$\begin{aligned} V_{AN} &= (37.28\angle -19.65^\circ)(12+j5) \\ &= (37.28\angle -19.65^\circ)(13\angle 22.62^\circ) \\ &= 484.64\angle 2.97^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_{AN} is, $484.64\angle 2.97^\circ \text{ V}$.

Step 6 of 7

Calculate the value of the voltage V_{BN} .

$$V_{BN} = I_b(12+j5)$$

Substitute $37.28\angle -139.65^\circ \text{ A}$ for I_b .

$$\begin{aligned} V_{BN} &= (37.28\angle -139.65^\circ)(12+j5) \\ &= (37.28\angle -139.65^\circ)(13\angle 22.62^\circ) \\ &= 484.64\angle -117.03^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_{BN} is, $484.64\angle -117.03^\circ \text{ V}$.

Step 7 of 7

Calculate the value of the voltage V_{CN} .

$$V_{CN} = I_c (12 + j5)$$

Substitute $37.28\angle 100.35^\circ$ A for I_c .

$$\begin{aligned} V_{CN} &= (37.28\angle 100.35^\circ)(12 + j5) \\ &= (37.28\angle 100.35^\circ)(13\angle 22.62^\circ) \\ &= 484.64\angle 122.97^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_{BN} is, $484.64\angle 122.97^\circ$ V .